microeconomics_2_L1

Subject: #microeconomics

Topic: Review

Definition. Let $F \geq 0$ denote fixed costs (independent of quantity q), and $VC(q) \geq 0$ denote the variable cost of producing quantity q.

> ✎ **Note**
>
> It is assumed that the function VC is continuous at $q = 0$ and $VC(0) = 0$.

Definition. The total cost of producing quantity q is given by:

$$C(q) = F + VC(q)$$

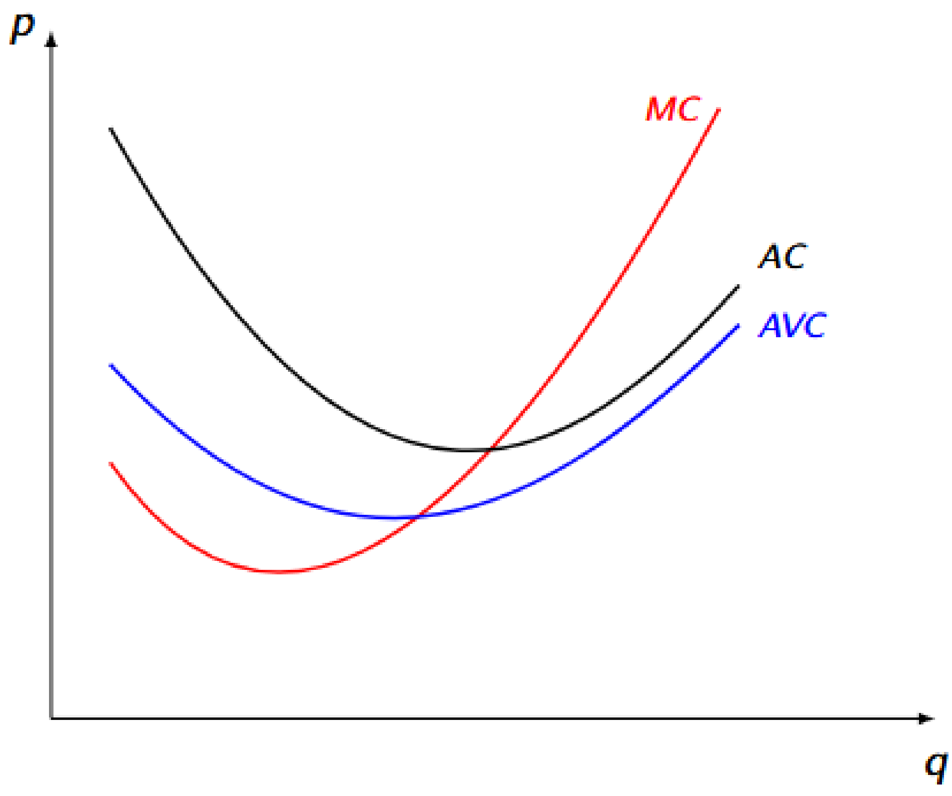
Definition. The marginal (incremental) cost for quantity q is:

$$MC(q) = C'(q) = VC'(q)$$

(When taking the derivative of the $C(q), F$ gets destroyed, as fixed costs are constant no matter how much is produced.)

Definition. The average costs (AC) and average variable costs (AVC) for quantity $q > 0$ are:

$$AC(q) = \frac{C(q)}{q} \quad \text{and} \quad AVC(q) = \frac{VC(q)}{q}$$



> **Note**
>
> The MC curve intersects the AC and AVC curves at their minimums (at interior points).

Definition. Fixed costs F can be further decomposed into **sunk costs** and **avoidable fixed costs.**

Definition. The total cost of producing quantity q is given by:

$$C(q) = \begin{cases} S & q = 0 \\ F + VC(q) & q > 0 \end{cases}$$

where:

- S is the sunk cost part of fixed costs ($S \leq F$)
- $F - S$ is the avoidable part of fixed costs

> **Note**
>
> Teachers sometimes assume that $S = F$.

> **Note**
>
> There will be an upcoming task where if you are unable to find the the fixed costs, it is recommended you draw the cost function $c(q)$.

Perfect Competition

Definition. A perfectly competitive market is an ideal (a benchmark) against which other models/markets are compared.

Main assumptions of perfect competition:

- Homogeneous good
- Perfectly divisible output
- Perfect information
- No transaction costs
- No externalities
- Price-taking behavior

> **Note**
>
> Perfectly competitive markets are rarely observed in the real world (if ever).

Definition. The firm's profit is:

$$\pi(q) = R(q) - C(q) = p \cdot q - C(q)$$

The firm maximizes its profit $\pi(q)$ by choosing $q \geq 0$.

The firm's decision process (a two-step process):

1. What positive output q maximizes profit?

2. Does the firm make enough profit to cover its costs, or is it better off by shutting down ($q = 0$)?

Step 1. To maximize profit when $q > 0$:

$$\max_{q>0} p \cdot q - C(q)$$

First-Order Condition (FOC) (interior optimum) — price equals marginal cost at $q^* > 0$:

$$p = MC(q^*)$$

Essentially, the FOC will check if the profit function will provide us with marginal profit as the firm continues producing more product.

The Second-Order Condition (SOC) must also be checked:

$$-MC'(q^*) \leq 0 \quad \Rightarrow \quad MC'(q^*) \geq 0$$

The SOC will ensure that not only the marginal costs ahead will be increasing, but that the marginal costs before the point are smaller than the current point.

Thus, intuitively, marginal costs must be increasing at q^*.

> **Warning**
>
> If at any point your $p^* < 0$, then you are doing something wrong.

Step 2. The shutdown condition: it must be checked that shutting down is suboptimal:

$$\pi(q^*) \geq \pi(0) \text{ (profit with p* vs profit when shutting down)}$$

$$p \cdot q^* - F - VC(q^*) \geq -S$$

$$p \geq AVC(q^*) + \frac{F - S}{q^*} \text{ (Remember that } \frac{VC}{q} = AVC)$$

or

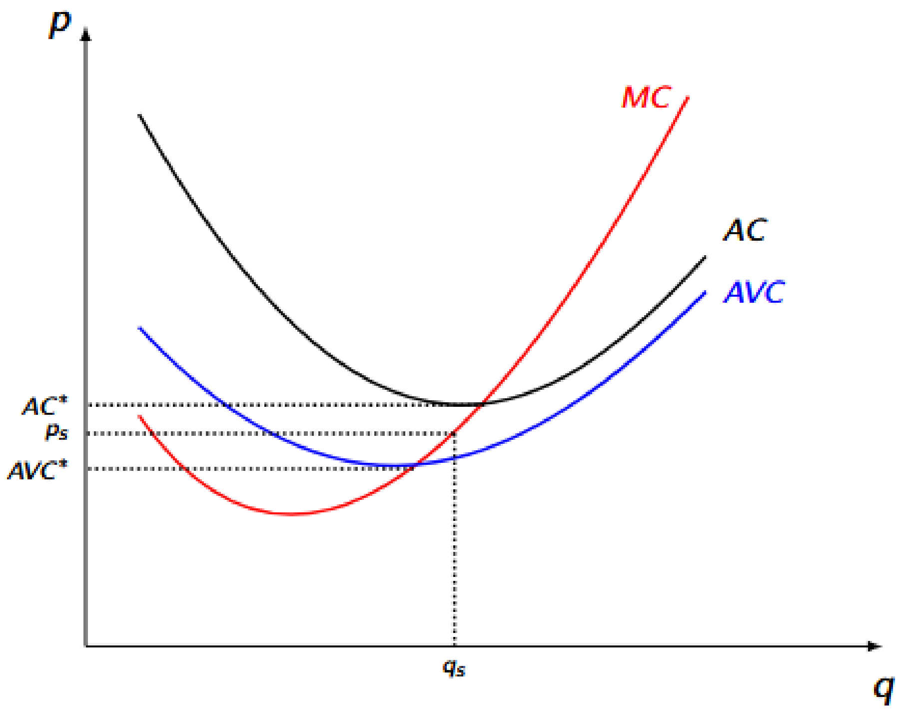
$$MC(q^*) \geq AVC(q^*) + \frac{F - S}{q^*}$$

Remember than the optimal price is the marginal cost of a product.

> **Note**
>
> The price is high enough to cover all avoidable costs.

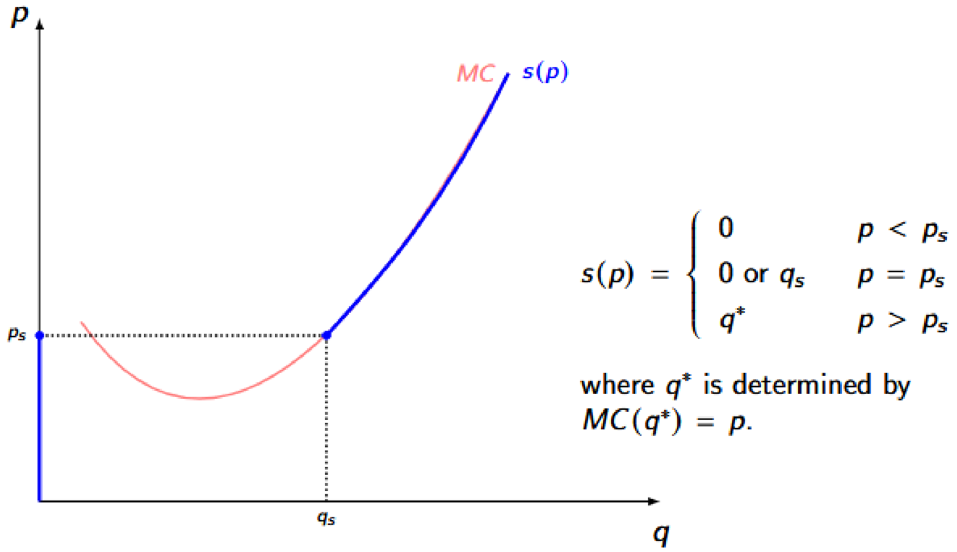
- if $S = F$ (all fixed costs are sunk), then $p \geq AVC(q^*)$
- if $S = 0$ (no sunk costs), then $p \geq AC(q^*)$

> **Note**
>
> You can solve for q_s and p_s using the following equations we got previously.
>
> $$MC(q_s) = AVC(q_s) + \frac{F - S}{q_s}$$
>
> $$p_s = MC(q_s)$$



The shutdown price p_s can be as low as AVC^* or as high as AC^*. Furthermore, if some of the shutdown costs are avoidable, then shutdown costs become higher.

A typical firm's supply - with an optimal output given any p.



Class Example 1. Suppose the firm's total cost of producing q is given by:

$$C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$$

Construct the firm's supply function $s(p)$.

Step 1: (finding q^*)

$$p = MC(q^*) = C(q)' \text{ (FOC)} \implies p = 2q \implies q^* = \frac{p}{2}$$

Step 2:

$$\pi(q^*) = p \cdot q^* - C(q^*) = 2q^{*2} - 16 - (q^*)^2 = (q^*)^2 - 16$$

$$\pi(q^*) \geq \pi(0)$$

$$q^{*^2} - 16 \geq -12 \implies (q^*)^2 \geq 4 \implies q^* = 2$$

As a result, we have:

$$S(p) = \begin{cases} 0 & p < 4; \\ 0 \text{ or } 2 & p = 0; \\ \frac{p}{2} & p > 4 \end{cases}$$

Review questions:

Cost Concepts

1. What are fixed costs?
2. What are variable costs?
3. What is assumed about the variable cost function $VC(q)$ at $q = 0$?
4. What is the total cost of producing quantity q?
5. What is the marginal cost (MC)?
6. Why does the derivative of $C(q)$ not include F when computing marginal cost?
7. What are average cost (AC) and average variable cost (AVC)?
8. Where does the MC curve intersect the AC and AVC curves?

Fixed, Sunk, and Avoidable Costs

9. Into what two categories can fixed costs F be decomposed?
10. How is the total cost function defined when distinguishing between sunk and avoidable costs?
11. What do S and $F - S$ represent?
12. What assumption do some teachers make about sunk costs S?

Perfect Competition

13. What is a perfectly competitive market?
14. List the main assumptions of a perfectly competitive market.

15. What is the profit function of a firm?

16. What are the two steps in the firm's decision process under perfect competition?
||1) Find the output q that maximizes profit.

17. Check whether producing yields higher profit than shutting down $(q = 0)$.||

Profit Maximization

17. Write the maximization problem for profit when $q > 0$.

18. What is the First-Order Condition (FOC) for profit maximization?

19. What does the Second-Order Condition (SOC) require?

Shutdown Condition

20. State the shutdown condition in terms of profits.

21. Rewrite the shutdown condition in terms of prices and costs.

22. Express the shutdown condition as a condition on price.

23. Express the shutdown condition using marginal cost.

24. What is the shutdown rule when $S = F$?

25. What is the shutdown rule when $S = 0$?

Sources: